



Empirical correlations for second virial coefficients of associated and quantum fluids covering a wide temperature range

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ABSTRACT

Characterizing interactions between two molecules, the second virial coefficient is of great value in calculating the effect of the real-fluid imperfection. The corresponding-state principle followed, the second-virial-coefficient correlation proposed in our previous work was extended to associated and quantum fluids. The effects of hydrogen bonds were carefully analyzed and the new associated polar terms gave physically reliable results for alcohols, amines, ammonia and water. Taking the reduced de Broglie wavelength as the characteristic parameter, a new corresponding-state treatment for quantum fluids was established. The new quantum term was applicable for all quantum fluids except for Neon and was in good agreement with the ab initio and experimental data when the reduced temperature is higher than 0.33. The second virial coefficients of Neon can be calculated by the simple spherical term formula. For the requirement of high-precision calculation, this paper also proposed the specific correlations of the second virial coefficients for He-3, He-4, Neon and water. Together with previous correlations for nonpolar and non-associated polar fluids, a second-virial-coefficient model for most common fluids in a wide temperature range was completed in this work.

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1. Introduction

Having a rigorous basis in statistical mechanics, virial coefficients are valuable in theoretical development of equilibrium thermodynamic properties for practical applications. The second virial coefficient characterizes the real-fluid imperfection caused by the interaction between two molecules. Only the second virial coefficient used, the truncated virial equation of state is reliable enough to calculate thermodynamic properties less than one third of the critical density in combination with the ideal-gas heat capacity. Therefore, a topic has been attracting much attention that how to predict second virial coefficients by the easily-accessible thermodynamic properties. Aside from the accuracy of that prediction, researchers made much effort on achieving the universality of the predictive method. The universality means the application for not only more kinds of fluids, but also a wider temperature range.

A number of generalized models for second virial coefficients were proposed in the literatures. The principles those models based on can be roughly divided into: (a) the corresponding-state principle (CS), (b) the chemical theory (CT), and (c) the group additivity principle (GA).

To figure out the universality of those generalized models, Table 1 gave the comparison of the merits and demerits for different methods. A second-virial-coefficient model could be applicable for a wide temperature range only when possessing four temperature-dependent characteristics [1]:

Condition A: $\lim_{T \rightarrow 0} B = -\infty$;

Condition B: $T \geq T_B, B \geq 0$;

Condition C: $\exists T', (\frac{dB}{dT})_{T < T'} > 0, (\frac{dB}{dT})_{T = T'} = 0, (\frac{dB}{dT})_{T > T'} < 0$;

Condition D: $\lim_{T \rightarrow +\infty} B = 0$.

Most correlations satisfied the conditions A and B while they usually cannot describe the reversal temperature (T_{rev}) or did not tend to be zero at extremely high temperatures. Also, a generalized model should cover most common fluids such as nonpolar, non-associated polar, associated polar and quantum fluids.

It could be seen from Table 1 that the corresponding-state method is the most developed one and shows the advantage of a relatively wide temperature range. However, few corresponding-state methods covered associated polar fluids and quantum fluids. Also, most corresponding-state methods gave invalid descriptions of second virial coefficients at high temperatures.

Therefore, models achieving the universality are the Nothnagel correlation [2], the Meng correlation [3] and Iglesias-Silva and Hall's correlation [4]. The Nothnagel correlation gave fluid-specific

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Table 1

The comparison of generalized second-virial-coefficient models.

Year	Author	Principle	Condition C	Condition D	Nonpolar	Non- associated polar	Associated polar	Quantum
1957	Pitzer and Curl [5]	CS	No	No	•			
1958	Black [6]	CS	No	No	•	•	•	
1966	Gunn et al. [7]	CS	No	No				•
1967	O'Connell and Prausnitz [8]	CS	No	No	•	•	•	
1969	Kreglewski [9]	CS	No	No	•	•	•	
1971	Halm and Stiel [10]	CS	No	No	•	•	•	
1973	Nothnagel et al. [2]	CT	No	No	•	•	•	•
1974	Tsonopoulos [11]	CS	No	No	•	•	•	
1975	Hayden and O'Connell [12]	CS	No	No	•	•	•	
1977	Tarakad and Danner [13]	CS	No	No	•	•	•	
1984	McCann and Danner [14]	GA	No	No	•	•	•	
1994	Weber [15]	CS	No	No	•	•	•	
1995	Virendra [16]	CS	No	No	•	•	•	
1999	Vetere [17]	CS	Yes	No	•	•	•	
2001	Iglesias-Silva and Hall [4]	CS	Yes	Yes	•	•	•	•
2004	Meng et al. [3]	CS	No	No	•	•	•	•
2016	Di Nicola et al. [18]	CS+GA	Yes	Yes	•	•		
2021	this work	CS	Yes	Yes	•	•	•	•

constants based on the experimental data for each fluid and was not a predictive method. The Meng correlation and Iglesias-Silva and Hall's correlation would be compared carefully in Section 3.

In our previous work [1], a new corresponding-state correlation model was proposed to describe the second virial coefficients of nonpolar and non-associated polar fluids covering an extremely wide temperature range. Based on the ab initio values and experimental data, those new correlations have a new functional form that can account for behaviors of fluids at extremely high temperatures. It is proven by the ab initio values that the new correlation could even give physically correct descriptions of second virial coefficients under extremely high temperature conditions up to 3000 times the critical temperature. Eq. (1) gives the second-virial-coefficient correlation:

$$B_r = \frac{Bp_c}{RT_c} = f^{(0)}(T_r) + \omega f^{(1)}(T_r) + f^{(2)}(T_r) \quad (1)$$

where, B_r is the reduced second virial coefficient, B is the second virial coefficient ($\text{cm}^3 \cdot \text{mol}^{-1}$), p_c is the critical pressure (MPa), R is the universal gas constant ($8.3144626 \text{ J} \cdot \text{mol}^{-1} \cdot \text{K}^{-1}$) [19], T_c is the critical temperature (K), $T_r = \frac{T}{T_c}$ is the reduced temperature, T is the temperature (K), ω is the acentric factor.

The simple spherical term ($f^{(0)}$), the nonpolar term ($f^{(1)}$) and the non-associated polar term ($f^{(2)}$) are determined as:

$$f^{(0)} = 119.5e^{-5.653T_r^{0.05}} - \frac{0.5925}{T_r^{0.7}} - \frac{0.1553}{T_r^{2.1}} - \frac{0.003901}{T_r^{4.7}} \quad (2)$$

$$f^{(1)} = 0.1818e^{-0.02324T_r} - \frac{0.06851}{T_r} - \frac{0.1562}{T_r^4} - \frac{0.0001456}{T_r^9} \quad (3)$$

$$f^{(2)} = \frac{a}{T_r^6} \quad (4)$$

for haloalkanes:

$$a = -3.486 \times 10^{-7} \mu_r^2 + 3.028 \times 10^{-11} \mu_r^4 - 1.391 \times 10^{-15} \mu_r^6 \quad (5)$$

and for other non-associated polar fluids:

$$a = -3.158 \times 10^{-6} \mu_r^2 + 1.035 \times 10^{-10} \mu_r^4 - 1.450 \times 10^{-15} \mu_r^6 \quad (6)$$

where μ_r is the reduced dipole moment defined as

$$\mu_r = \frac{\mu^2 p_c}{1.01325 T_c^2}, \quad (7)$$

where μ is the dipole moment in Debye.

The basis of those new correlations is the corresponding-state principle. Different fluids would follow the same physical law when adopting appropriate reducing parameters. Take simple fluids (Argon, Krypton and Xenon) for example, their second virial coefficients can be described by a simple reduced correlation, $f^{(0)}$. However, as that principle is to find molecular commonalities, it is not strictly correct when molecular differences can not be ignored any more. Those three simple fluids show discrepancies at extremely low and high temperatures, for instance, the spread of their reduced second virial coefficients is about 3% at $T_r = 17$ [1]. But the corresponding-state principle is still a powerful theory in characterizing common features of fluids.

For nonpolar fluids, their reduced second virial coefficients would always be lower than simple fluids at low temperatures and higher at high temperatures. Thus, the first two terms of the Eq. (1), $B_r = f^{(0)} + \omega f^{(1)}$, can account for behaviors of various kinds of nonpolar fluids. That success is resulted from the acentric factor proposed by Pitzer [5,20] in 1955. "This third parameter is required because the intermolecular force in complex molecules is a sum of interactions between various parts of the molecules—not just their centers—hence the name acentric factor is suggested [20]." The three-parameter corresponding-state theorem is an undoubtedly excellent idea. The soundness and reliability of that method were proved by massive ab initio and experimental data of second virial coefficients for nonpolar fluids.

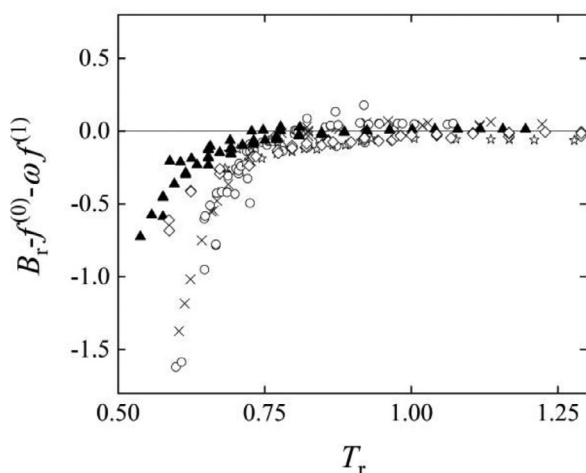
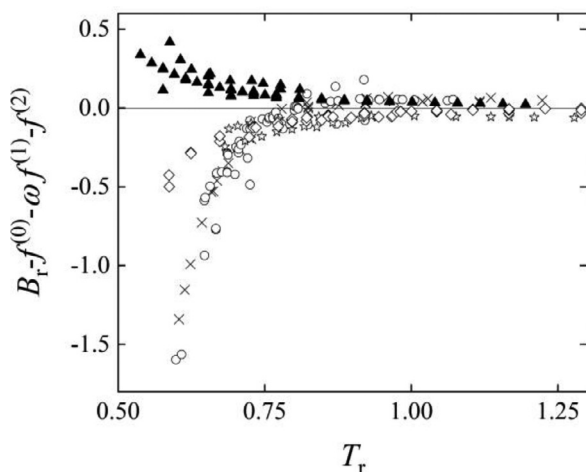
Polar fluids have lower second virial coefficients than nonpolar fluids with the same acentric factor at low temperatures. Thus, the polar term ($f^{(2)}$) ought to be added in their descriptions. For polar fluids, the dipole-dipole attractions are resulted by the uneven distribution of electrons. That uneven distribution is usually due to atoms' different affinity for electrons. O'Connell and Prausnitz [8] introduced the reduced dipole moment to measure the effect of the polarity in 1967. And that is the fourth characteristic parameter in the generalized second-virial-coefficient correlation. But studies showed that the haloalkanes confirm to the different curve from other non-associated fluids when μ_r is used as the characteristic parameter [1,3]. That is consistent with Tsonopoulos's conclusion: "However, the reduced dipole moment is a satisfactory correlating parameter only when each family of compounds is considered separately [11]." Therefore, the quantitative representation of their second virial coefficients should be achieved by the proper classification of polar fluids.

As a consequence of the corresponding-state principle, Eq. (1) has a solid theoretical basis that makes itself successful in the description of second virial coefficients for nonpolar and

Table 2

Thermophysical properties [56] and deviations of experimental data from Eq. (1) for associated fluids.

Full name	Short Formula	T_c /K	p_c /MPa	ω	μ_r	RMSD/cm ³ ·mol ⁻¹
Methanol	CH ₄ O	513.38	8.216	0.5625	88.91	6127
Ethanol	C ₂ H ₆ O	514.71	6.268	0.646	66.76	224.9
1-Propanol [57,58]	C ₃ H ₈ O	536.8	5.169	0.6204	49.90	52.61
2-Propanol [24,57,59]	C ₃ H ₈ O	508.3	4.764	0.6669	50.20	44.80
1-Butanol [57]	C ₄ H ₁₀ O	563	4.414	0.5895	37.91	149.49
2-Butanol [57]	C ₄ H ₁₀ O	536.2	4.202	0.5768	39.79	41.90
Tert-butanol [57]	C ₄ H ₁₀ O	506.2	3.972	0.6152	42.66	51.98
2-Methyl-1-propanol [57]	C ₄ H ₁₀ O	547.8	4.295	0.5857	37.99	53.93
Methylamine	CH ₅ N	430.05	7.46	0.2814	68.33	5.399
Ethylamine	C ₂ H ₇ N	456.15	5.62	0.2848	39.69	20.16
Dimethylamine [39,47]	C ₂ H ₇ N	437.2	5.34	0.2999	29.32	11.60
Trimethylamine [39,47,60]	C ₃ H ₉ N	433.25	4.07	0.2062	8.004	16.59
Diethylamine [31,37,47]	C ₄ H ₁₁ N	496.6	3.71	0.3039	12.58	50.72
Ammonia	NH ₃	405.56	11.3634	0.256	147.3	53.24
Water	H ₂ O	647.096	22.064	0.3443	178.9	22.83

**Fig. 1.** The relationships between $B_r - f^{(0)} - \omega f^{(1)}$ and T_r of the experimental data [21] for (x) methanol, (o) ethanol, (v) methylamine, (d) ammonia and (Δ) water.**Fig. 2.** The relationships between $B_r - f^{(0)} - \omega f^{(1)} - f^{(2)}_{\text{other nonassociated fluids}}$ and T_r of the experimental data [21] for (x) methanol, (o) ethanol, (v) methylamine, (d) ammonia and (Δ) water.

polar fluids. The aim of this article is to extend this correlation to associated and quantum fluids.

Among molecules with moieties that hydrogen atoms covalently bonded to other electronegative atoms (–FH, –NH, or –OH), hydrogen bonds are formed between hydrogen atoms and elec-

tronegative atoms. And fluids containing hydrogen bonds are associated fluids. Stronger than the dipole-dipole attractions among non-associated polar molecules, this dimerization usually results in lower second virial coefficients than non-associated polar fluids at low temperatures. Similar to non-associated polar fluids, μ_r can be used as the characteristic parameter to measure the effect of hydrogen bonds. Also, the associated fluids should be divided into different families and then can be described in the framework of Eq. (1).

Quantum fluids are also called light fluids due to low molecular weight and simple molecular structure (one or two atoms). Exhibiting quantum mechanical effects at the macroscopic scale, quantum fluids have certain anomalous thermodynamic properties. Take the acentric factor for example, all quantum fluids have negative acentric factors which are definitely different from Pitzer's physical explanation. Therefore, the terms of $\omega f^{(1)}$ and $f^{(2)}$ in Eq. (1) should be excluded. Also, the characteristic parameter to measure quantum effects is a key to the new corresponding-state treatment for quantum fluids.

2. Associated fluids

To explore the hydrogen-bond effects on second virial coefficients, it should be figured out firstly that the differences of B between associated fluids and nonpolar fluids with the same acentric factor. Fig. 1 gives the relationships between $B_r - f^{(0)} - \omega f^{(1)}$ and T_r for methanol, ethanol, methylamine, ammonia and water. Similar to non-associated polar fluids, the associated fluids have much lower B than nonpolar fluids at low temperatures.

Describing the effect of polarity, the reduced dipole moment is used as the characteristic parameter to compare the quantitative relationships of the second virial coefficients for associated polar fluids and non-associated polar fluids. Fig. 2 presents the relationships between $B_r - f^{(0)} - \omega f^{(1)} - f^{(2)}_{\text{other nonassociated fluids}}$ and T_r for methanol, ethanol, methylamine, ammonia and water. It can be observed that the associated polar fluids have lower B than non-associated polar fluids at the same reduced temperatures and pressures. Those low-temperature negative deviations mean that the associated polar fluids have smaller specific volume than the non-associated polar fluids. It can be physically explained that the hydrogen bonds between or within associated molecules are stronger than the dipole-dipole attractions between non-associated polar molecules.

As molecular differences are too large among polar fluids, associated polar fluids should be carefully classified into small groups that molecules in one group should have similar structures. The associated polar terms were determined as:

$$f^{(2)} = \mu_r \vartheta \quad (8)$$

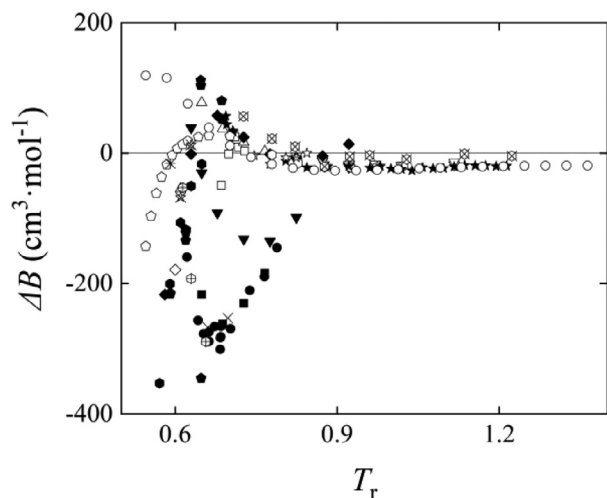


Fig. 3. Deviations of the experimental data from Eq. (1) for methanol: (□) Eucken and Meyer [22]; (●) Lambert et al. [23]; (Δ) Kretschmer and Wiebe [24]; (▼) Bottomley and Spurling [25]; (▽) Kell and McLaurin [26]; (◆) Kudchadker and Eubank [27]; (○) Lalzadecrabtree et al. [28]; (●) Noppe et al. [29]; (★) Bich et al. [30]; (☆) Olf et al. [31]; (○) Kerl and Varchmin [32]; (✱) Boyes et al. [33]; (○) Massucci et al. [34]; (■) Rowlinson [35]; (×) Lambert et al. [36]; (▲) Everett and Munn [37]; (▲) Dēdek [38]; (◇) Millen and Mines [39]; (◁) Pathak [40]; (►) Ohgaki et al. [41]; (▷) Barhala and Marchidan [42]; (●) Bich et al. [43].

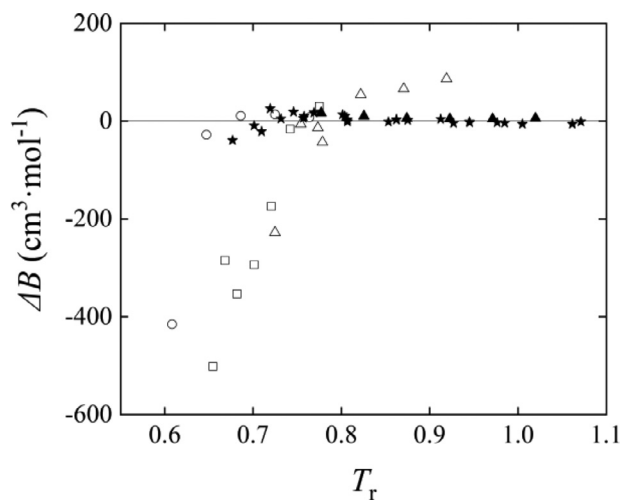


Fig. 4. Deviations of the experimental data from Eq. (1) for ethanol: (□) Lambert [44]; (○) Kretschmer and Wiebe [24]; (▲) Noppe et al. [29]; (Δ) Wilson et al. [45]; (★) Bich et al. [46].

For alcohols:

$$\vartheta = \frac{0.002541}{T_r^4} - \frac{0.001682}{T_r^6} \quad (9)$$

for amines:

$$\vartheta = \frac{-0.0007694}{T_r^4} \quad (10)$$

for the ammonia:

$$\vartheta = \frac{-0.0001734}{T_r^6} \quad (11)$$

Figs. 3–7 compare the experimental data with present work for methanol, ethanol, methylamine, ethylamine and ammonia. It can be observed that the experimental data for those associated fluids are in great discrepancy. Present associated polar terms give physically reliable results for alcohols, amines and ammonia. Table 2 lists the comparison of the associated fluids' experimental data with the present work.

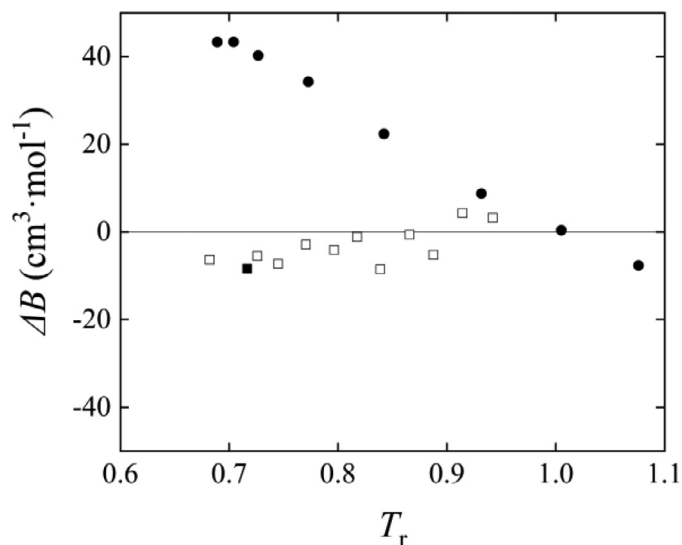


Fig. 5. Deviations of the experimental data from Eq. (1) for methylamine: (□) Lambert and Strong [47]; (■) Millen and Mines [39]; (●) Adam et al. [48].

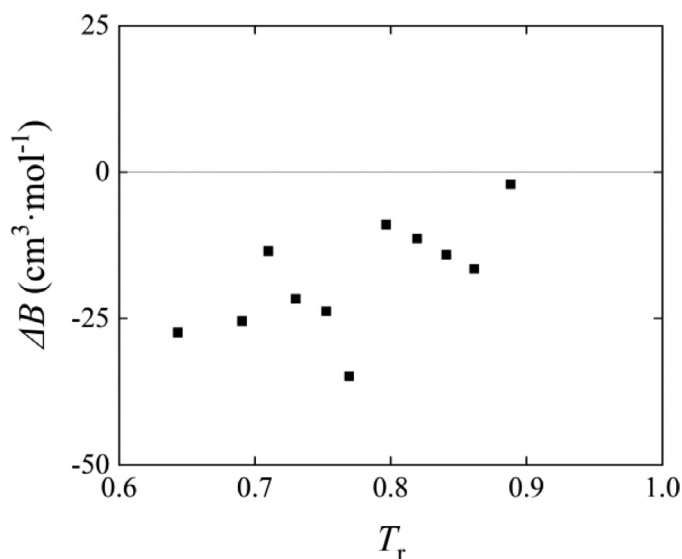


Fig. 6. Deviations of the experimental data from Eq. (1) for ethylamine: (■) Lambert and Strong [47].

Water is the only associated fluid in Fig. 2 that has positive deviations for $T_r < 0.75$, which looks a little strange. As a matter of fact, the particularity of water can be attributed to its hydrogen-bond effect: the hydrogen-bond effect is not only reflected in the reduced dipole moment, but also caused an excessive acentric factor. The acentric factor of water is 0.3443, which is greater than 0.300 of n-hexane and equivalent to 0.349 of n-heptane. However, water has a simple molecular structure with only 3 atoms, far less than the 23 atoms of n-heptane. It is obviously impossible to be explained by what Pitzer said for the acentric factor, "This third parameter is required because the intermolecular force in complex molecules is a sum of interactions between various parts of the molecules—not just their centers". One reasonable explanation is that the strong hydrogen-bond effect greatly reduces the vapor pressure, resulting in the excessive acentric factor. Thus, the $\omega f^{(1)}$ for water already contains the hydrogen-bond effect to a large extent, and only the residual part needs to be described by the associated polar term. Of course, it is practicable to give a separate associated polar term for water. Fig. 8 presents the relation-

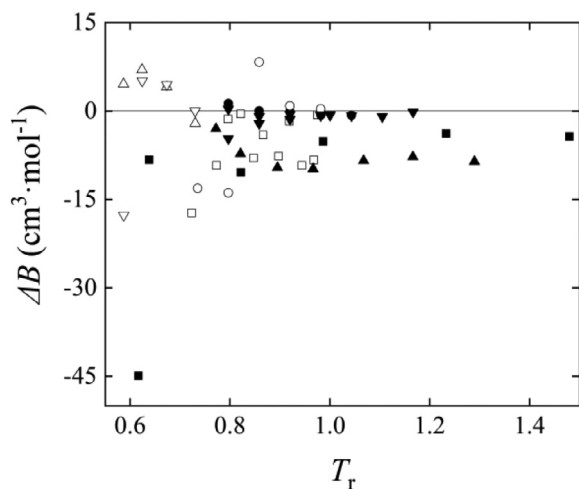


Fig. 7. Deviations of the experimental data from Eq. (1) for ammonia: (■) Stockmayer [49]; (□) Lambert and Strong [47]; (●) Cheh et al. [50]; (○) Date [51]; (▲) Adam and Schramm [52]; (▽) Elias et al. [53]; (▼) Glowka [54]; (∇) Schramm et al. [55].

ship between $B_r - f^{(0)} - \omega f^{(1)}$ and T_r for the ab initio values and the calculated values of IAPWS95 and Harvey-Lemmon correlation. The $B_r - f^{(0)} - \omega f^{(1)}$ increases to be positive from $T_r = 0.87$ and then decreases to be negative for $T_r > 1.62$. There is a clear fact that the $B_r - f^{(0)} - \omega f^{(1)}$ would come to zero at extremely high temperatures. Therefore, it is difficult to accurately describe this trend by a simple-form associated polar term.

The present associated polar term for water is determined as:

$$\vartheta = -\frac{0.001341}{T_r} + \frac{0.003082}{T_r^2} - \frac{0.001654}{T_r^3} - \frac{0.000004978}{T_r^9} \quad (12)$$

Fig. 9 shows that this associated polar term is in good agreement with the experimental data, the ab initio values and the calculated values of the Harvey-Lemmon correlation and IAPWS95.

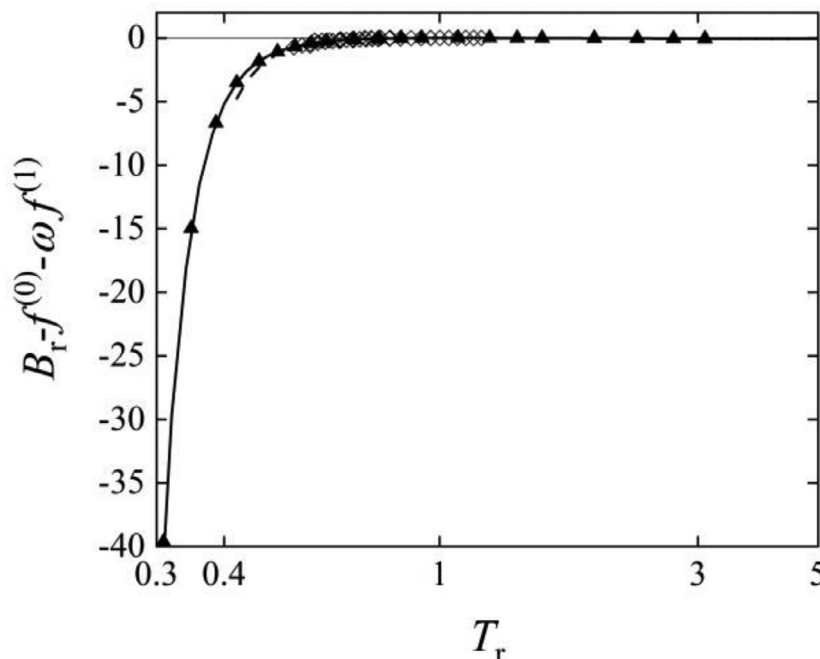


Fig. 8. Deviations of the experimental data from $B_r - f^{(0)} - \omega f^{(1)}$ for water: (▲) ab initio values [61]; (○) experimental data [21]; (—) calculated values of IAPWS95 [62]; (---) calculated values of Harvey-Lemmon correlation [63]; (-·-) calculated values of $\mu_r \cdot \text{Eq. (12)}$.

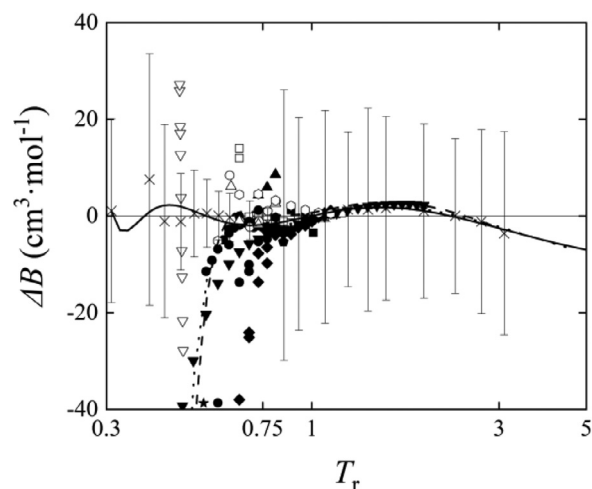


Fig. 9. Deviations of the ab initio values, the experimental data and the calculated values of Eq. (13) from Eq. (1) for water: (×) ab initio values [61]; (■) Abdulagatov et al. [64]; (□) Kell et al. [65]; (●) Eubank et al. [66]; (○) Noppe et al. [29]; (▲) Skripka [67]; (△) Noppe [29]; (▼) Le Fevre et al. [68]; (∇) Treiner et al. [69]; (◆) Kell et al. [70]; (○) Kell et al. [71]; (●) Collins and Keyes [72]; (★) McCullough et al. [73]; (☆) Bohlander and Gebbie [74]; (○) Lotter et al. [75]; (—) calculated values of IAPWS95 [62]; (---) calculated values of Harvey-Lemmon correlation [63]; (-·-) calculated values of Eq. (13).

A specific correlation out of the framework of Eq. (1) was more convenient to account for water's unique behaviors:

$$B_{r,\text{water}} = 0.05914e^{0.1303T_r^{0.5}} - \frac{0.2802}{T_r^{1.6}} - \frac{0.1204}{T_r^4} - \frac{0.0009290}{T_r^9}. \quad (13)$$

Compared to water's experimental data, the temperature range and the uncertainty of the ab initio values do not show much superiority, especially at high temperatures. Therefore, the experimental data and the Harvey-Lemmon correlation play an important

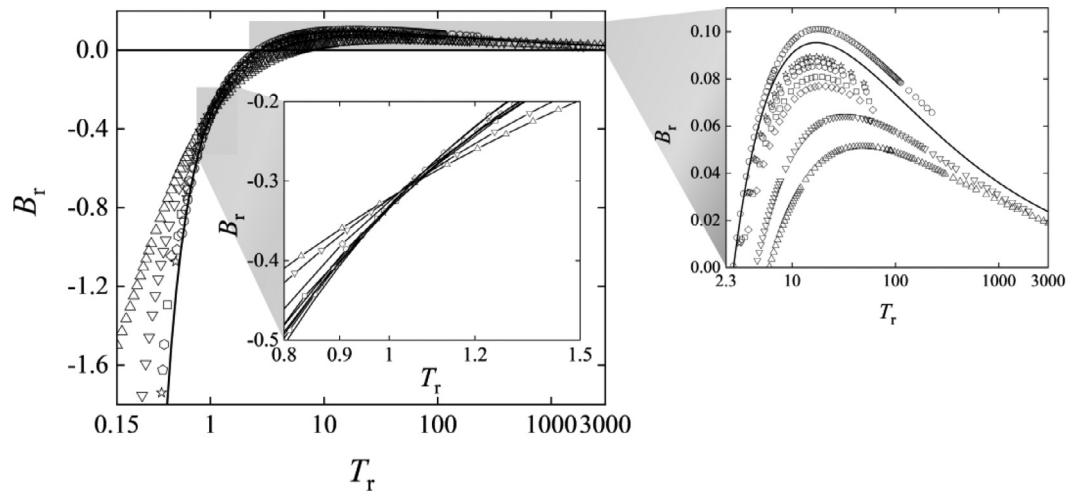


Fig. 10. The $B_r - T_r$ relationships of ab initio values for (○) Neon [78]; (△) He-3 [79,80]; (▽) He-4[79, 80]; (◇) H₂ [77]; (◇) D₂ [77]; (☆) T₂ [77]; (○) DT [77]; (□) HD [77]; (—) the calculated values of Eq. (2).

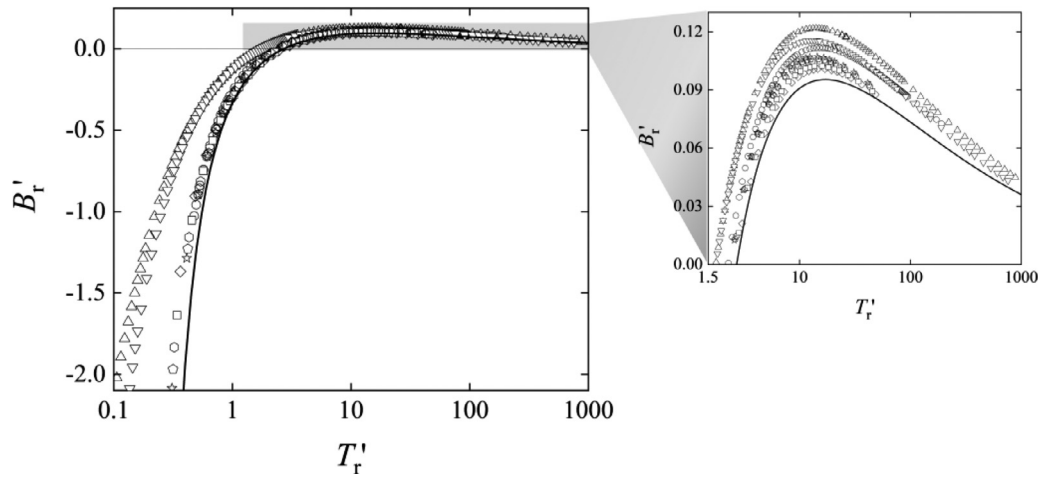


Fig. 11. The $B'_{r'} = \frac{B(p_c + 0.8106)}{R(T_c + 8)} - T'_{r'} = \frac{T}{T_c + 8}$ relationships of ab initio values for (○) Neon [78]; (△) He-3 [79,80]; (▽) He-4[79, 80]; (◇) H₂ [77]; (◇) D₂ [77]; (☆) T₂ [77]; (○) DT [77]; (□) HD [77]; (—) the calculated values of Eq. (2).

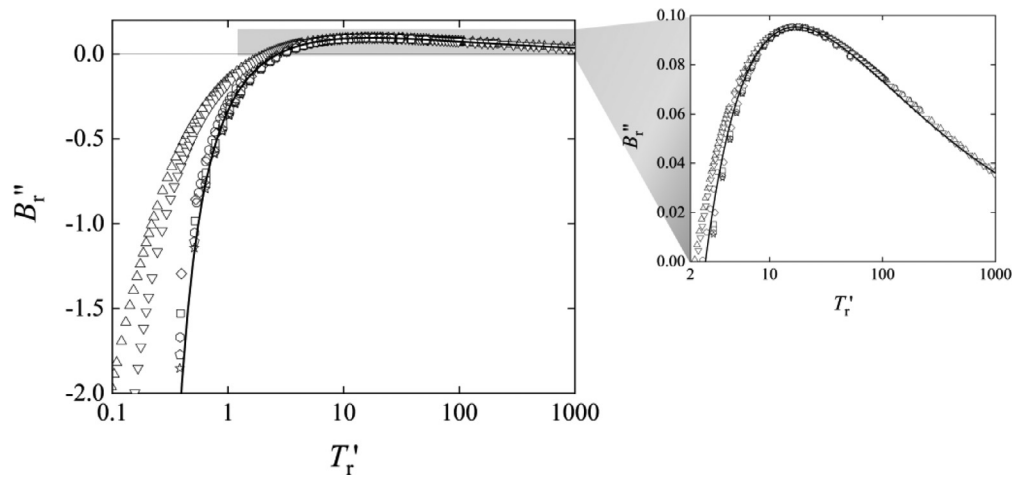
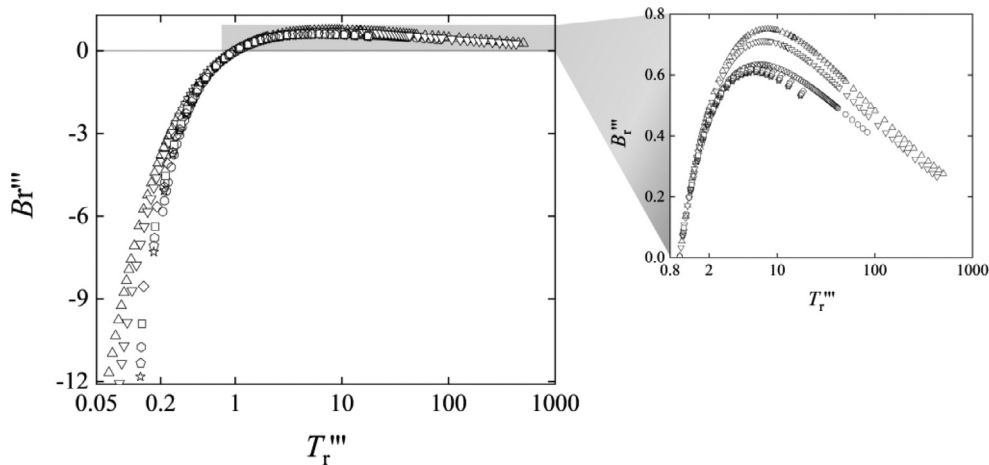
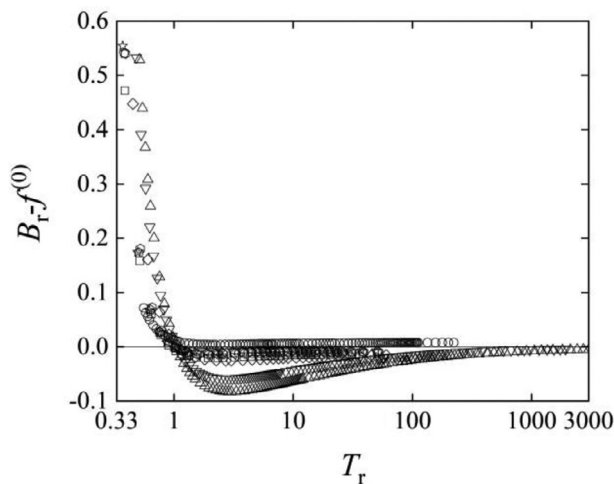
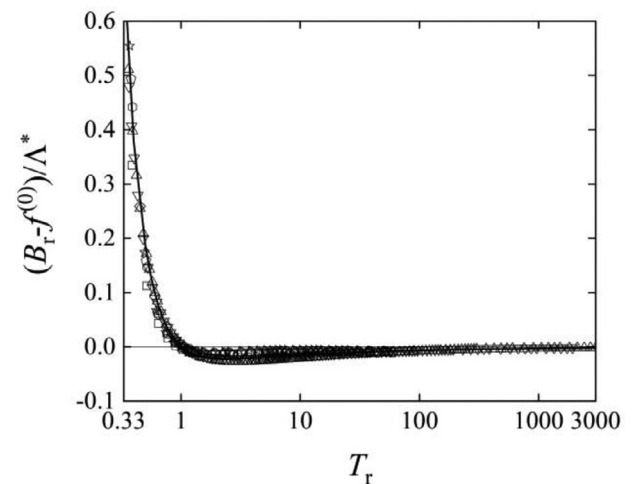


Fig. 12. The $B''_{r'} = \frac{B(p_c + k_2)}{R(T_c + k_1)} - T'_{r'} = \frac{T}{T_c + k_1}$ relationships of ab initio values for (○) Neon [78]; (△) He-3 [79,80]; (▽) He-4[79, 80]; (◇) H₂ [77]; (◇) D₂ [77]; (☆) T₂ [77]; (○) DT [77]; (□) HD [77]; (—) the calculated values of Eq. (2).

Table 3

Thermophysical properties [56,76] and deviations of experimental data (exp) or ab initio data (ab) from Eq. (15) or specific correlations (spe) for quantum fluids.

Short formula	Λ^*	T_c /K	p_c /MPa	ω	k_1 /K	k_2 /MPa	T_B /K	v_B /cm ³ ·mol ⁻¹	RMSD /cm ³ ·mol ⁻¹	RMSD _{spe} /cm ³ ·mol ⁻¹
He-3	3.08	3.3157	0.114604	-0.4723	5.789	0.478	19.879	16.494	4.018 ^{exp} (2.826) ^{ab}	28.822 ^{exp} (0.314) ^{ab}
He-4	2.67	5.1953	0.22761	-0.385	4.767	0.415	23.195	17.134	3.050 ^{exp} (2.041) ^{ab}	4.451 ^{exp} (0.162) ^{ab}
H ₂	1.729	33.145	1.2964	-0.219	4.799	0.563	108.285	25.988	6.224 ^{exp} (1.675) ^{ab}	
HD [77]	1.412	35.91	1.484	-0.18	2.679	0.441	112.329	26.266	(1.696) ^{ab}	
D ₂	1.223	38.34	1.6796	-0.136	0.497	0.22	114.122	26.402	4.766 ^{exp} (2.310) ^{ab}	
DT [77]	1.095	39.42	1.77	-0.13	-0.420	0.139	115.431	26.550	(2.640) ^{ab}	
T ₂	1	40.44	1.861	-0.14	-1.307	0.056	116.155	26.588	2.139 ^{exp} (3.132) ^{ab}	
Ne	0.593	44.492	2.6786	-0.0387	2.211	-0.115	119.516	22.026		9.111 ^{exp} (0.009) ^{ab}

**Fig. 13.** The $B_r''' = \frac{B}{v_B} - T_r''' = \frac{T}{T_B}$ relationships of ab initio values for (○) Neon [78]; (△) He-3 [79,80]; (▽) He-4[79, 80]; (◇) H₂ [77]; (○) D₂ [77]; (☆) T₂ [77]; (○) DT [77]; (□) HD [77].**Fig. 14.** The $B_r - f^{(0)}$ - T_r relationships of ab initio values for (○) Neon [78]; (△) He-3 [79,80]; (▽) He-4[79, 80]; (◇) H₂ [77]; (○) D₂ [77]; (☆) T₂ [77]; (○) DT [77]; (□) HD [77].**Fig. 15.** The $\frac{B_r - f^{(0)}}{\Lambda^*}$ - T_r relationships of ab initio values for (△) He-3 [79,80]; (▽) He-4[79, 80]; (◇) H₂ [77]; (○) D₂ [77]; (☆) T₂ [77]; (○) DT [77]; (□) HD [77] and (—) Eq. (16).

role in the fitting for $T_r > 1.15$. The present specific correlation is in good agreement with all those data as shown in Fig. 9.

3. Quantum fluids

Quantum fluids have negative acentric factors as shown in Table 3. Those negative acentric factors do not conform to the physical explanation when Pitzer proposed this parameter, and are just values calculated from the vapor pressure affected by quantum effects. Thus, Eq. (1) is not appropriate for quantum fluids, at least $\omega f^{(1)}$ is not appropriate. However, the simple spherical term

is still a necessary standard for comparison. It can be considered to be mainly resulted from quantum effects that there exist differences between the second virial coefficients of quantum fluids and the calculated values of $f^{(0)}$.

To explore their second-virial-coefficient differences caused by quantum effects, all quantum fluids' ab initio data were compared with the simple spherical term ($f^{(0)}$) in Fig. 10. Except for Neon, most quantum fluids have higher second virial coefficients than the simple spherical term at low temperatures and lower second virial coefficients at high temperatures. It can be observed that their curves intersect with $f^{(0)}$ at about $T_r = 1.05$ except

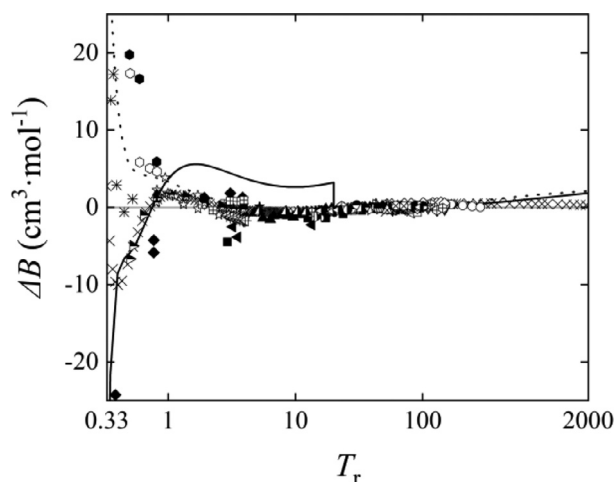


Fig. 16. Deviations of the ab initio values and experimental data for He-4 from Eq. (15) for $T_r > 0.33$: (×) ab initio values [79,80]; (■) Holborn and Otto [83]; (◁) Van Agt et al. [84]; (▴) Gibby et al. [85]; (▵) Tanner and Masson [86]; (●) Wiebe et al. [87]; (○) Keesom and Kraak [88]; (□) Keesom and Walstra [89]; (◻) Michels and Wouters [90]; (+) Edwards and Roseveare [91]; (✱) Kistemaker and Keesom [92]; (⊞) Keesom and Walstra [93]; (•) Schneider and Duffie [94]; (◊) Yntema and Schneider [95]; (⊕) Cottrell and Hamilton [96]; (⊗) Kramer and Miller [97]; (⊙) Beenakker et al. [98]; (⊖) Silberberg et al. [99]; (⊗) Verekamp and Beenakker [100]; (▲) White et al. [101]; (⊗) Canfield et al. [102]; (⊕) Witonsky and Miller [103]; (Δ) Hoover et al. [104]; (▽) Kalfoglou and Miller [105]; (△) Suh and Storvick [106]; (⊗) Sullivan and Sonntag [107]; (◆) Boyd et al. [108]; (⊕) Hall and Canfield [109]; (⊗) Kudchadker and Eubank [27]; (⊞) Provine and Canfield [110]; (●) Berry [111]; (▲) Vogl and Hall [112]; (▽) Vanadurongwan et al. [113]; (◊) Kell et al. [114]; (⊙) Prasad and Kudchadker [115]; (◁) Berry [116]; (▴) Skripka [67]; (⊙) Guban and Michel [117]; (★) Holste et al. [118]; (★) Sengers and Hastings [119]; (⊙) Mansoorian et al. [120]; (⊙) Barkan [121]; (⊞) Schramm and Buchner [122]; (●) Dunlop et al. [123]; (⊙) Kemp et al. [124]; (⊙) Mallu et al. [125]; (★) Karnatsevich et al. [126]; (▲) Patel [127]; (▽) Schramm et al. [55]; (◁) Zhang et al. [128]; (⊙) Qian et al. [129]; (⊙) Duarte et al. [130]; (—) calculated values of Meng correlation; (...) calculated values of Iglesias-silva and Hall's correlation.

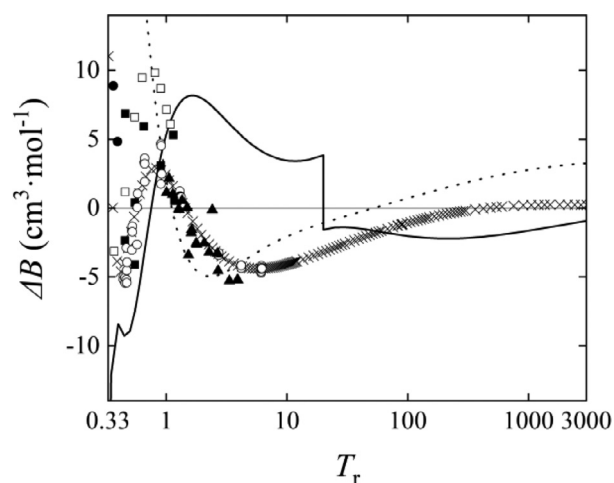


Fig. 17. Deviations of the ab initio values and experimental data for He-3 from Eq. (15) for $T_r > 0.33$: (×) ab initio values [79,80]; (■) Keller [131]; (□) Grimrud and Wernitz [132]; (●) Cameron and Seidel [133]; (◊) Maticotta et al. [134]; (▲) Karnatsevich and Bogoyavlenskii [126]; (—) calculated values of Meng correlation; (...) calculated values of Iglesias-silva and Hall's correlation.

for Neon. Also, different from simple fluids, quantum fluids do not have fixed reduced Boyle temperature ($T_{r,B}$) and reduced reversed temperature ($T_{r,rev}$). The $T_{r,B}$ and $T_{r,rev}$ of D_2 , DT and T_2 are close to those of simple fluids, while the $T_{r,B}$ and $T_{r,rev}$ of He-3, He-4, H_2 and HD are higher than those of simple fluids.

Among all those physical differences, changes of $T_{r,B}$ and $T_{r,rev}$ make it necessary to develop a new corresponding-state treat-

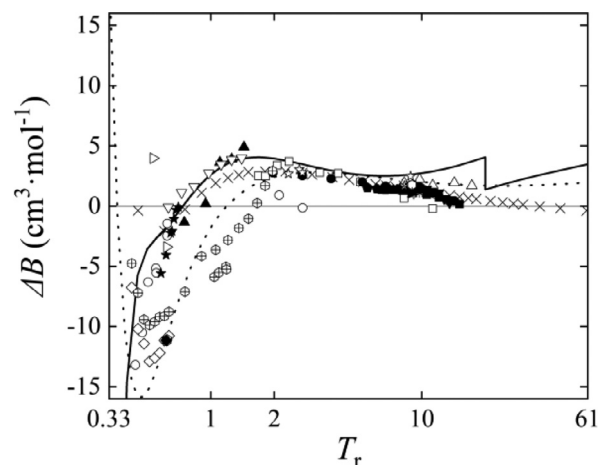


Fig. 18. Deviations of the ab initio values and experimental data for H_2 from Eq. (15) for $T_r > 0.33$: (×) ab initio values [77]; (□) Kamerlingh Onnes and Braak [135]; (⊗) de Haas [136]; (●) Holborn [83]; (◊) Van Agt and Kamerlingh Onnes [84]; (✱) Verschroye [137]; (▲) Nijhoff and Keesom [138]; (Δ) Bartlett et al. [139]; (▽) Gibby et al. [85]; (▽) Long and Brown [140]; (▴) Edwards and Roseveare [91]; (+) Cottrell et al. [96]; (◆) Beenakker et al. [98]; (◊) Verekamp and Beenakker [100]; (c) Knaap et al. [141]; (★) El Hadi et al. [142]; (✱) Mihara et al. [143]; (○) Lazalde-Crabtree et al. [144]; (■) Perez et al. [145]; (★) Schramm et al. [146]; (⊕) Dunlop et al. [123]; (✱) Schramm et al. [55]; (—) calculated values of Meng correlation; (...) calculated values of Iglesias-silva and Hall's correlation.

ment for quantum fluids. Since scholars put forward many ideas to describe the thermodynamic properties of quantum fluids, their methods can be first applied in the treatment for second virial coefficients of quantum fluids:

In Newton's work of activity coefficients [81], the behaviors of quantum fluids were tried to be uniformized by adding constants to critical parameters. That means the reduced parameters are re-defined as $T_{r'} = T/(T_c + 8)$ and $p_{r'} = p/(p_c + 0.8106)$ (Newton defined $p_{r'} = p/(p_c + 8)$ in the pressure unit of atm, while the pressure unit is MPa in this work). As shown in Fig. 11, that method was tried in the application of second virial coefficients that $T_{r'} = T/(T_c + 8)$ and $B_{r'} = B(p_c + 0.8106)/R(T_c + 8)$. Although better conformity is achieved in the high-temperature region, their new reduced second virial coefficients show great discrepancies at low temperatures.

Thus, one possibility is considered in this work that the behaviors of quantum fluids may be uniformized by adding fluid-specific constants to critical parameters. Quantum fluids were made the same $T_{r,rev}$ and $B_{r,rev}$ as those of the simple spherical term, which means $T_{r,rev}' = T_{rev}/(T_c + k_1) = 17.1277$ and $B_{r,rev}' = B_{rev}(p_c + k_2)/R(T_c + k_1) = 0.09534$. The constants added to T_c and p_c are given in Table 3. However, Fig. 12 shows that this trial makes the high-temperature trends of helium isotopes different from those of hydrogen isotopes, and their new reduced second virial coefficients show great discrepancies at low temperatures as well.

Based on Newton's method and Pitzer-Curl-type correlation [5], Gunn [7] made the 'effective' critical parameters dependent on the temperature, the molar mass and the characteristic constants. However, take the 'effective' critical temperature ($T_c^{Gunn} = T_c/(1 + c_1/mT)$) for example, c_1 is negative for $T_{r,rev}^{Gunn} = T_{rev}/T_c^{Gunn} = 17.1277$. It means the reduced temperature is negative for $T < -c_1/m$. Therefore, this method may not be an appropriate corresponding-state treatment for second virial coefficients of quantum fluids.

Iglesias-silva and Hall [4] took the method different from the Pitzer-Curl-type correlation. Instead of critical parameters, they used the Boyle temperature as the normalizing variable for the temperature. They proposed a generalized second-virial-coefficient

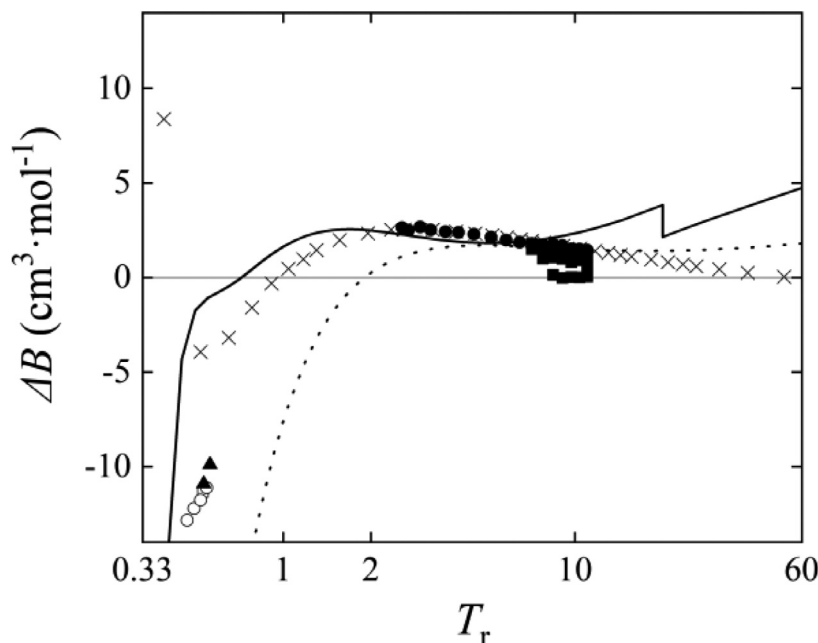


Fig. 19. Deviations of the ab initio values and experimental data for D_2 from Eq. (15) for $T_r > 0.33$: (x) ab initio values [77]; (■) Michels and Goudek [147]; (□) Beenakker et al. [98]; (●) Michels et al. [148]; (○) Varekamp and Beenakker [100]; (▲) Knaap et al. [141]; (—) calculated values of Meng correlation; (---) calculated values of Iglesias-silva and Hall's correlation.

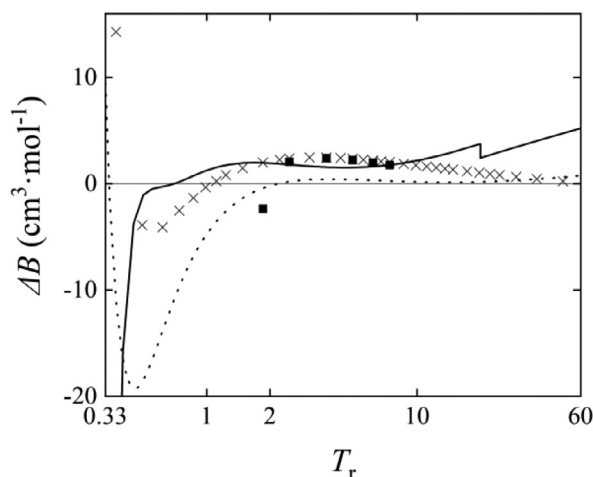


Fig. 20. Deviations of ab initio values and experimental data for T_2 from Eq. (15) for $T_r > 0.33$: (x) ab initio values [77]; (■) Mills et al. [149]; (—) calculated values of Meng correlation; (---) calculated values of Iglesias-silva and Hall's correlation.

model in the form of a product of several exponential terms. Iglesias-silva and Hall's correlation gave impressively accurate description of high-temperature B for quantum fluids.

Therefore, the other possibility was considered in this work that the critical parameters of quantum fluids are not reliable enough to be the reducing parameters (especially for the critical parameters of hydrogen isotopes), and that the Boyle parameters may be better choices. Also, the critical properties are influenced by many-body forces while B characterizes the real-fluid imperfection caused by the interaction between two molecules. So, their Boyle parameters were calculated from the ab initio second virial coefficients, and tried as the reducing parameters as shown in Fig. 13. The Boyle temperature T_B is defined as the temperature at which B becomes zero and the Boyle volume is defined as $v_B = T_B \left(\frac{dB}{dT} \right)_{T=T_B}$. The reduced parameters are redefined as $T_r''' = \frac{T}{T_B}$ and $B_r''' = \frac{B}{v_B}$.

This method can not achieve the low-temperature conformity as well, although the high-temperature consistency is improved.

Meng et al. [82] proposed a piecewise function for quantum fluids to improve the accuracy. But that correlation was not continuous at $T_r = 20$. Thus, this method was not adopted in this work.

It is conventional to use the reduced De Broglie wavelength as the characteristic parameter to measure quantum effects. The reduced De Broglie wavelength is defined as

$$\Lambda^* = h / \sigma \sqrt{m\epsilon} \quad (14)$$

where h is Planck's constant, m is the molecular mass, σ and ϵ are the intermolecular potential parameters.

As shown in Fig. 10, except for Neon, when Λ^* is used as the characteristic parameter, the second-virial-coefficient differences of quantum fluids from $f^{(0)}$ can be summarized as: (a) quantum fluids with larger Λ^* have higher second virial coefficients at low temperatures and lower second virial coefficients at high temperatures; (b) quantum fluids with larger Λ^* have gentler increase of B at low temperatures and gentler decline at high temperatures; and (c) quantum fluids have larger $T_{r,B}$ and $T_{r,rev}$ as their Λ^* becomes larger.

Presenting their ab initio relationships between $B_r - f^{(0)}$ and T_r , Fig. 14 shows that all quantum fluids have the similar trend except for neon. As analyzed in the previous paragraph, the reduced De Broglie wavelength is a propriate characteristic parameter. Fig. 15 compare the ab initio $\frac{B_r - f^{(0)}}{\Lambda^*} - T_r$ relationships for all quantum fluids except for Neon. Although their $\frac{B_r - f^{(0)}}{\Lambda^*} - T_r$ relationships do not strictly conform the same curve, their differences are reduced largely. Then $\frac{B_r - f^{(0)}}{(\Lambda^*)^n}$ was also tried to achieve better conformance. However, it was found that although $n = 1.5$ can make the consistency better for $T_r > 1$, it will also make the consistency worse for $T_r < 0.7$. As the magnitude of B_r at low temperatures is much larger than that at high temperatures, the optimum n of $\frac{B_r - f^{(0)}}{(\Lambda^*)^n}$ was determined as 1 because it achieved the best low-temperature conformance among helium isotopes and hydrogen isotopes.

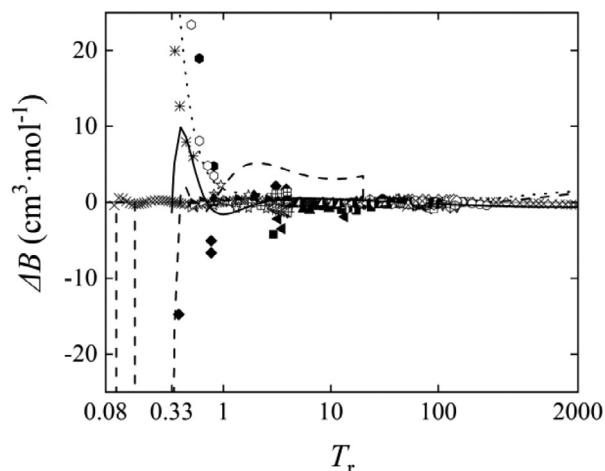


Fig. 21. Deviations of ab initio values and experimental data for He-4 from the Eq. (17): (×) ab initio values [79,80]; (■) Holborn and Otto [83]; (●) Van Agt et al. [84]; (◐) Gibby et al. [85]; (◑) Tanner and Masson [86]; (◒) Wiebe et al. [87]; (◓) Keesom and Kraak [88]; (◔) Keesom and Walstra [89]; (◕) Michels and Wouters [90]; (+) Edwards and Roseveare [91]; (◖) Kistemaker and Keesom [92]; (◗) Keesom and Walstra [93]; (◘) Schneider and Duffie [94]; (◙) Yntema and Schneider [95]; (◚) Cottrell and Hamilton [96]; (◛) Kramer and Miller [97]; (◜) Beenakker et al. [98]; (◝) Silberberg et al. [99]; (◞) Varekamp and Beenakker [100]; (▲) White et al. [101]; (◟) Canfield et al. [102]; (◠) Witonsky and Miller [103]; (◡) Hoover et al. [104]; (◢) Kalfoglou and Miller [105]; (◣) Suh and Storvick [106]; (◤) Sullivan and Sonntag [107]; (◥) Boyd et al. [108]; (◦) Hall and Canfield [109]; (◧) Kudchadker and Eubank [27]; (◨) Provine and Canfield [110]; (◩) Berry [111]; (◪) Vogl and Hall [112]; (◫) Vanadurongwan et al. [113]; (◬) Kell et al. [114]; (◭) Prasad and Kudchadker [115]; (◮) Berry [116]; (◯) Skripka [67]; (◰) Guban and Michel [117]; (◱) Holste et al. [118]; (◲) Sengers and Hastings [119]; (◳) Mansoorian et al. [120]; (◴) Barkan [121]; (◵) Schramm and Buchner [122]; (◶) Dunlop et al. [123]; (◷) Kemp et al. [124]; (◸) Mallu et al. [125]; (◹) Karnatsevich et al. [126]; (◺) Patel [127]; (◻) Schramm et al. [55]; (◼) Zhang et al. [128]; (◽) Qian et al. [129]; (◾) Duarte et al. [130]; (—) calculated values of Meng correlation; (...) calculated values of Iglesias-silva and Hall's correlation, (---) calculated values of Eq. (15).

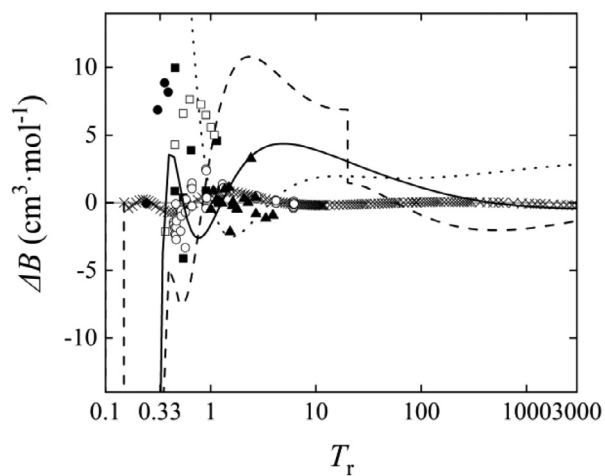


Fig. 22. Deviations of ab initio values and experimental data for He-3 from the Eq. (18): (×) ab initio values [79,80]; (■) Keller [131]; (□) Grimsrud and Wertz [132]; (●) Cameron and Seidel [133]; (◐) Maticotta et al. [134]; (▲) Karnatsevich and Bogoyavlenskii [126]; (—) calculated values of Meng correlation; (...) calculated values of Iglesias-silva and Hall's correlation, (---) calculated values of Eq. (15).

Therefore, the second virial coefficients of quantum fluids can be described by:

$$B_r = f^{(0)} + \Lambda^* f^{(q)} \quad (15)$$

A quantum term ($f^{(q)}$) was added to the simple spherical term directly. Also, because the critical parameters of hydrogen isotopes are not as reliable as those of hydrogen, the ab initio values of hydrogen, He-3 and He-4 were weighed mostly in the fitting of the

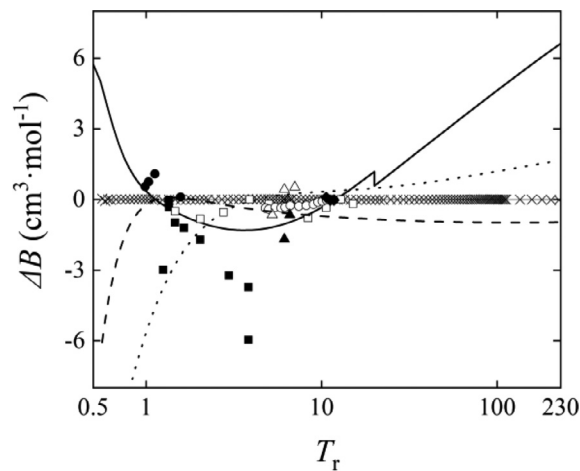


Fig. 23. Deviations of ab initio values and experimental data for Neon from present work: (×) ab initio [78]; (■) Crommelin et al. [151]; (□) Holborn and Otto [83]; (●) Gibbons [152]; (◐) Schmiedel et al. [153]; (▲) Onnes and Crommelin [154]; (△) Vogl and Hall [112]; (◐) Skripka [67]; (◑) Kerl [155]; (—) calculated values of Meng correlation; (...) calculated values of Iglesias-silva and Hall's correlation, (---) calculated values of Eq. (2).

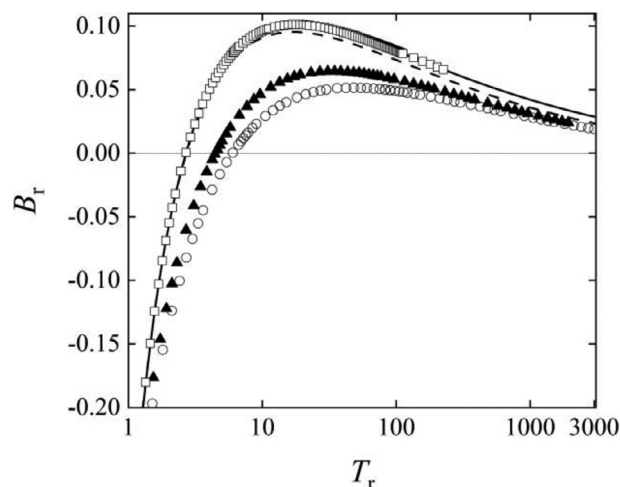


Fig. 24. The $B_r - T_r$ relationships of (—) the calculated values of Eq. (2), (---) Eq. (15) as well as the ab initio values of (◐) He-3 [79,80], (▲) He-4 [79,80] and (□) Neon [78].

quantum term:

$$f^{(q)} = -\frac{0.03566}{T_r^{0.37}} + \frac{0.03619}{T_r^{2.7}} \quad (16)$$

Because the simple spherical term is fitted by the ab initio values of Ar, Kr and Xe and their minimum reduced temperature of their available ab initio data is 0.334, Eq. (15) is applicable for all quantum fluids apart from Neon for $T_r > 0.33$. The minimum temperature to which Eq. (15) can be applied is 1.1 K for He-3 and 1.6 K for He-4.

Figs. 16–20 compare the ab initio values, the experimental data, the Meng correlation, the Iglesias-silva and Hall's correlation with Eq. (15) for He-3, He-4, hydrogen, deuterium and tritium, respectively. It is worth mentioning that even though Iglesias-silva and Hall did not use the critical temperature as the reducing parameter, the reduced temperature ($T_r = T/T_c$) was taken as the abscissa for convenient comparison. Present correlations are in good agreement with those quantum fluids for $T_r > 0.33$. Iglesias-silva and Hall's work is in good agreement with those quantum fluids at

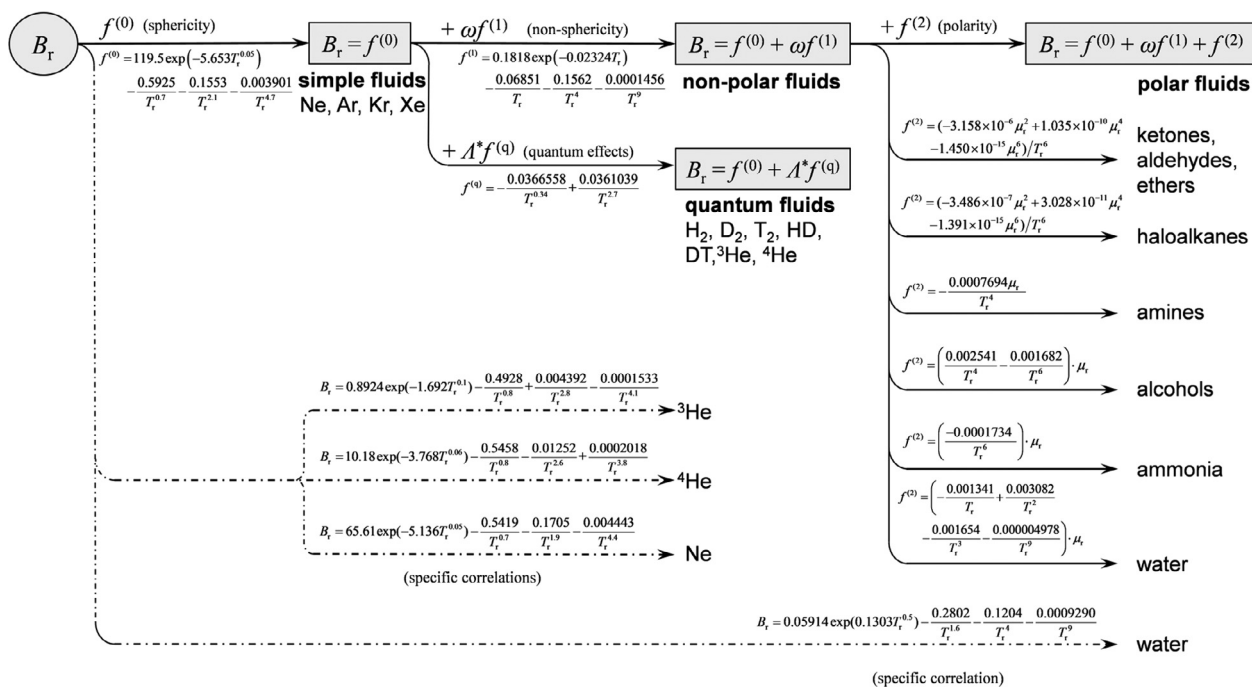


Fig. 25. A generalized framework of the second virial coefficients for all fluids.

high temperatures but often presents large deviations for $T_r < 1$. Table 3 gives the comparison of quantum fluids with present work.

Based on the already excellent potential for helium, Czachorowski et al. [80] improved the pair potential and gave the state-of-art virial coefficients with $T_{\min} = 0.5$ K ($T_{r,\min} = 0.151$ for Helium-3 and $T_{r,\min} = 0.096$ for Helium-4). The accuracy and reliability of those ab initio values were also proven by Song and his colleagues [150] that their calculated data were in great consistency. The $T_{r,\min}$ of Helium-3 and Helium-4 is lower than the $T_{r,\min}$ of the simple spherical term ($T_{r,\min} = 0.33$). Thus, it is necessary to give correlations describing the whole ab initio temperature range of those typical quantum fluids. For Helium-4:

$$B_{r,\text{He-4}} = 10.18 e^{-3.768 T_r^{0.06}} - \frac{0.5458}{T_r^{0.8}} - \frac{0.01252}{T_r^{2.6}} + \frac{0.0002018}{T_r^{3.8}} \quad (17)$$

for Helium-3:

$$B_{r,\text{He-3}} = 0.8924 e^{-1.692 T_r^{0.1}} - \frac{0.4928}{T_r^{0.8}} + \frac{0.004392}{T_r^{2.8}} - \frac{0.0001533}{T_r^{4.1}} \quad (18)$$

The applicable temperature range of those correlations is (0.5–10,000) K. Figs. 21 and 22 compare the ab initio values, the experimental data, the Meng correlation, the Iglesias-silva and Hall's correlation with Eqs. (17) and (18) for He-4 and He-3, separately. The ab initio $\Delta B_{\max}$ is $0.596 \text{ cm}^3 \cdot \text{mol}^{-1}$ at $T_r = 0.106$ for Helium-4 and $0.845 \text{ cm}^3 \cdot \text{mol}^{-1}$ at $T_r = 1.357$ for Helium-3.

Neon has the biggest molar mass among all quantum fluids, and shows a different trend from other quantum fluids as shown in Fig. 10. Its second virial coefficients are very close to but a little bit higher than Krypton in the whole temperature range. Therefore, the simple spherical term is in good agreement with its ab initio values as shown in Fig. 23. A specific correlation is proposed for Neon for more accurate descriptions. Similar to the functional form of the simple spherical term, neon has a second-virial-coefficient correlation as:

$$B_{r,\text{Ne}} = 65.61 e^{-5.136 T_r^{0.05}} - \frac{0.5419}{T_r^{0.7}} - \frac{0.1705}{T_r^{1.9}} - \frac{0.004443}{T_r^{4.4}} \quad (19)$$

Fig. 23 compares the ab initio values, the experimental data, the Meng correlation, the Iglesias-silva and Hall's correlation with Eq. (19). The ab initio $\Delta B_{\max}$ is $0.03027 \text{ cm}^3 \cdot \text{mol}^{-1}$ at $T_r = 0.562$. Also, Fig. 24 shows that the calculated values of this correlation are always higher than those of the simple spherical term. It proves that this correlation can even give reliable predictions at extremely high temperatures.

Fig. 25 gave a summary of the previous and the present work. The second virial coefficients for most common fluids can be calculated within this framework through the proper classification of fluids. The generalized correlations were shown along the solid lines for simple, nonpolar and polar fluids, and the specific correlations were shown along the dotted lines for He-3, He-4, Neon and water.

4. Conclusions

As the second virial coefficient characterizes interactions between two molecules, the effect of the real-fluid imperfection can be accurately calculated by using the second virial coefficient for less than one third of the reduced density. The corresponding-state principle adopted, the common laws followed by second virial coefficients of different fluids can be found through appropriate dimensionless methods and characteristic parameters.

The corresponding-state correlations proposed by Xu et al. was extended to associated and quantum fluids in this work.

For associated fluids, the effects of hydrogen bonds were carefully analyzed. The new associated polar terms gave physically reliable results for alcohols, amines, ammonia and water.

For quantum fluids, a new corresponding-state treatment was proposed in this work. The new quantum term was applicable for all quantum fluids except for Neon, and deviations from ab initio values were no more than $15 \text{ cm}^3 \cdot \text{mol}^{-1}$ for $T_r > 0.33$.

Neon has a different trend but can be described by the simple spherical term. For convenient use or higher degree of accuracy, specific correlations were proposed for water, He-3, He-4 and Neon.

The extended correlations gave reliable descriptions for associated and quantum fluids in a wide temperature range. Therefore, a generalized second-virial-coefficient model was completed in this work with high accuracy and universality. The new model can apply for not only more kinds of fluids, but also a much wider temperature range.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Liu Xu: Conceptualization, Methodology, Formal analysis, Writing – original draft, Visualization. **Hang-Tao Liu:** Methodology. **Zhen Yang:** Supervision. **Yuan-Yuan Duan:** Conceptualization, Resources, Software, Writing – review & editing, Supervision, Project administration, Funding acquisition.

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